KOUSHAL KUMAR REDDY CHAGARI

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EDUCATION

University of Texas at Austin Aug 24 - Present

GPA: 3.95/4 | *M.S & Ph.D. in ECE, Wireless Networking and Communications Group (WNCG)*

Advisors: Prof. Jeffrey Andrews & Prof. Kaushik Chowdhury

Indian Institute of Technology Madras Nov 20 - Jul 24

Bachelor of Technology (Honors) in Electrical Engineering

CGPA: 9.82/10 | Department Rank : **1**

PUBLICATIONS

[1] **K. K. Chagari**, S. Akoum, K. Chowdhury, and J. G. Andrews, "Physics-Informed Generative Modeling for FR3 Channels Based on Experimental Measurements," Submitted to IEEE ICC 2027.

[2] **K. K. Chagari**, K. Nieman, O. Kanhere, and S. Akoum, "Environment Reconstruction Using 6G ISAC Measurements," Submitted to IEEE ICC 2027.

[3] O. Kanhere, K. Nieman, **K. K. Chagari**, and S. Akoum, "A Resource Allocation Framework for CP-OFDM ISAC Systems," Submitted to IEEE ICC 2027.

[4] **K. K. Reddy Chagari**, K. Chowdhury, S. Akoum, and J. G. Andrews, "Generative Modeling for FR3 Channels based on Experimental Measurements," in *2025 59th Asilomar Conference on Signals, Systems, and Computers*, 2025, pp. 1183–1187. DOI: 10.1109/IEEECONF67917.2025.11443799.

PATENTS

[1] O. Kanhere, **K. K. R. Chagari**, K. Nieman, S. Ghassemzadeh, and S. Akoum, "Multi-Stage Detection and Classification Systems and Methods for Use in Integrated and Sensing Communication Networks," U.S. Patent applied, Mar. 24, 2026.

[2] K. Nieman, **K. K. R. Chagari**, O. Kanhere, and S. Akoum, "Bistatic Doppler Filtering Method for Rejection of Multi-Bounce Reflections," U.S. Patent to apply soon, 2026.

3GPP CONTRIBUTION

- Performed Rel-20 6G ISAC evaluation on sensing-assisted communication (SAC) for AT&T's 3GPP contributions

SCHOLASTIC ACHIEVEMENTS

- Awarded the **Cockrell Graduate School Fellowship** for being among the best PhD applicants to UT Austin
- Awarded the **Siemens Prize** for the graduating B.Tech student with best academic record in Electrical Engineering
- Awarded the **Summer@EPFL** Fellowship, among a cohort of 50 students from across the world
- Awarded **Ramasarma V Kolluri Memorial Prize** for the highest CGPA in the third year of Electrical Engineering
- Awarded **B Jayant Baliga Scholarship** twice for the best academic record in 2nd, 3rd years of Electrical Engineering
- Awarded **Sri V Rajagopalan Memorial Prize** for the highest CGPA in the second year of Electrical Engineering
- Awarded **M Shankaraiah and M Sarada Scholarship** for the best academic performance in my sophomore year
- Awarded **Sri K Krishnamurthi Award** for the second-highest CGPA in the academic year 2020-21
- Awarded **Notional Award** for being among the top 7% of students admitted to IIT Madras on the basis of JEE rank
- Nominated for **OPJEMS Scholarship** twice in 2021 and 2022 for academic and leadership excellence
- Secured All India Rank **327** out of 1,50,000 students in JEE Advanced 2020
- Secured **100 percentile** and achieved All India Rank **22** out of more than 14 lakh students in JEE Mains 2020
- Secured **4th** rank in the state out of more than 2.5 lakh students in TS EAMCET 2020
- Secured **5th** rank in the state out of more than 2.5 lakh students in AP EAMCET 2020
- Secured **International Rank 3** in NSO conducted by Science Olympiad Foundation in the year 2019-20
- Qualified for Kishore Vaigyanik Protsahan Yojana (**KVPY**) **scholarship** in both years, 2019-20 and 2018-19

Research Projects

- **Generative Models for FR3 Channels & Digital Twin Validation** — Aug 24-Present

(Advisor: Dr. Jeffrey Andrews & Dr. Kaushik Chowdhury, WNCG, 6G@UT, AT&T Labs, UT Austin)

  - Developed an **MPC-domain** physics-grounded generative modeling framework for real-world FR3 measurements
  - Outperformed channel-domain generative modeling by 60% and off-the-shelf **Sionna OSM** ray tracing by 71%

- **Bachelor's Thesis - Broadcast Positioning System using ATSC 3.0** — Jan 24-Jun 24

(Advisor: Dr. Giridhar K, Telecommunication and Wireless Sensing - TelWiSe Group, IIT Madras)

  - Developed an **alternate PNT system** to GPS, leveraging HPHT broadcast towers for bandlimited ATSC systems
  - Simulated a hybrid algorithm based on matched filtering and **ESPRIT** to achieve RMSE of < 0.6 m

- **Summer@EPFL Research Internship - Radar synchronization** — May 23 - Aug 23

(Advisor: Dr. Haitham Al Hassanieh, Sensing and Networking Systems - SENS Lab, EPFL)

  - Synchronized two AWR2243 radars to achieve **coherent processing** and enable a larger virtual array
  - Customized PCB design of AWR2243 single-chip radar to include connectors required for finer clock synchronization

- **Automatic CCC-CG LNA Circuit Synthesis in TSMC 65 nm Process** — Aug 22 - May 23

(Advisor: Dr. S. Aniruddhan, Dept. of Electrical Engineering, IIT Madras)

- **Design of Low Voltage Subsystem Module - Formula SAE** — May 21 - Feb 22

(Advisor: Dr. Satyanarayanan Seshadri, Dept. of Applied Mechanics, IIT Madras)

Industry Experience

- **Environment Sensing Using 6G ISAC, AT&T Labs, Austin** — Jun 26 - Aug 26

  - Developed bistatic Doppler filtering for multi-bounce rejection and RF point cloud processing pipeline
  - Augmented RF mesh with Google Maps mesh to demonstrate continually updatable geospatial digital twins

- **ISAC Analysis Internship, AT&T Labs, Austin** — Jun 25 - Aug 25

  - Developed detection and tracking algorithms for ISAC measurements in the FR3 band at 14.5 GHz
  - Explored OFDM resource allocation trade-offs in terms of bandwidth, sampling rate, duty cycle for sensing accuracy

- **Embedded Software Developer - Curneu MedTech Pvt Ltd** — Dec 21 - Jan 22

  - Designed an accurate **real-life training system** for neurosurgeons performing brain aneurysm operations

Courses Completed

- Adaptive Signal Processing
- Multirate DSP
- Digital Communication Systems
- Communication Networks
- Wireless Communication
- Probability & Stochastic Processes
- Convex Optimization
- Statistical Machine Learning
- Diffusion Models
- Mobile Communication Networks & Systems
- Advanced Wireless & Radar Systems

Teaching Experience

- TA for ECE 313H: Signals and Systems Honors class for Fall 2024 by Prof. Jeffrey Andrews at UT Austin
- TA for NPTEL MOOC course Multirate DSP taken by Prof. David Koilpillai in Jan-May 2024 at IIT Madras
- TA for course EE3110: Probability Foundations for Electrical Engineers in the Jul-Nov 2023 semester at IIT Madras